Name: ____________________________ (   ) CT Group: 24S0 _______

RAFFLES INSTITUTION

2024 YEAR 6 TERM 3 TIMED PRACTICE

27 June 2024

H2 PHYSICS

Section B

INSTRUCTIONS TO CANDIDATES

Write your name, index number and CT Group.
Write your answers to Section B in the spaces provided on the question paper.

You are advised to write all your workings and answers clearly. Marks may be deducted for unclear workings.

For Examiner's Use		
Section A	MCQ	/ 30
Section B	1	/ 10
	2	/ 9
	3	/ 9
	4	/ 8
	5	/ 10
	6	/ 10
	7	/ 9
	8	/ 15
Deductions		
Total		/ 110

This document consists of **21** printed pages.

Section B (80 marks)

1 Coherent light of a single wavelength is incident normally on a double slit as shown in Fig. 1.1.

Both slits in the double-slit arrangement are rectangular, each with a width of 0.10 mm. The separation of the two slits is 0.60 mm.

A very large screen is placed parallel to the plane of the double slit at a large distance *D* from the slits.

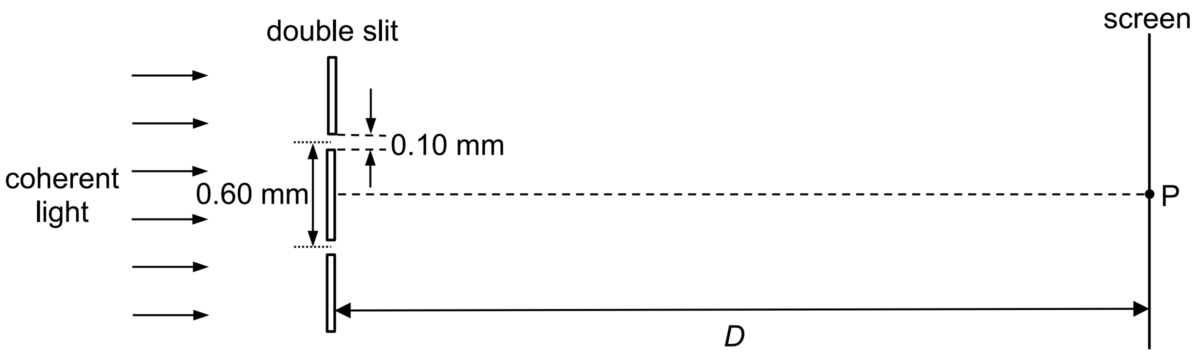

Fig. 1.1 (not to scale)

The centre of the interference pattern formed on the screen is at point P.

The variation with position of the intensity of the light on part of the screen is shown in Fig. 1.2.

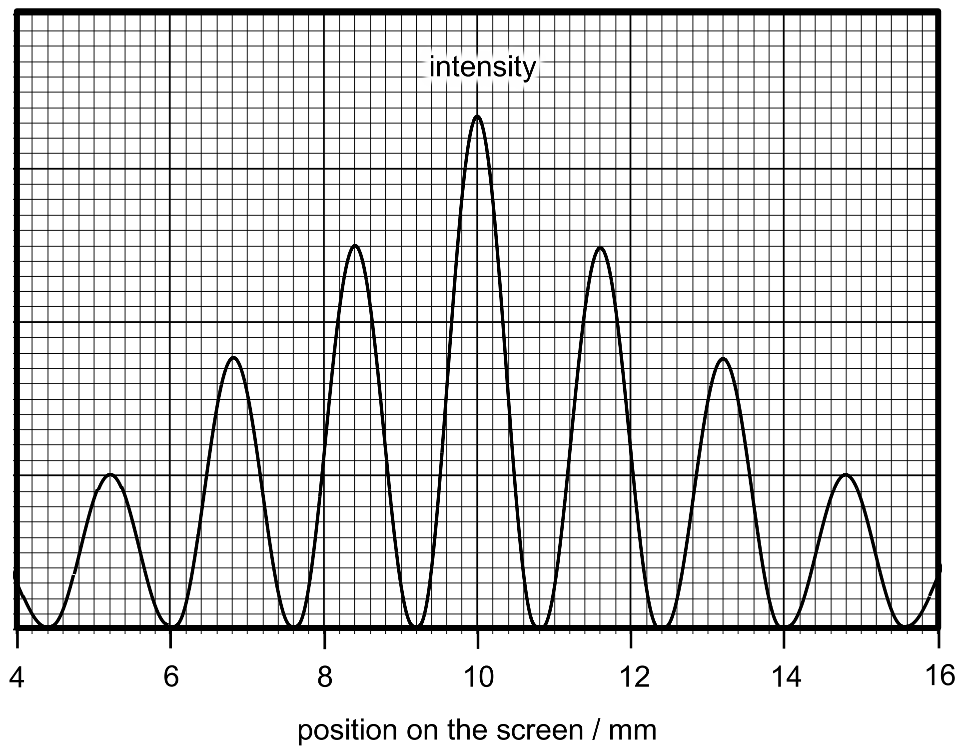

Fig. 1.2

(a) Explain the formation of the maximum and minimum intensity fringes of the interference pattern.

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.... [2]

(b) Calculate the width of the central fringe of the single-slit diffraction pattern seen on the screen if one of the slits is covered.

Show your working clearly.

fringe width = ... mm [2]

(c) Each of the following changes is made separately.

Describe how the appearance of the fringes of the interference pattern on the screen will compare to the interference pattern before the changes were made.

(i) The width of each slit is halved while keeping the slit separation the same.

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.... [3]

(ii) The wavelength of the coherent light incident on the slits is doubled.

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.... [3]

2 **(a)** Explain how molecular movement gives rise to pressure exerted by a gas.

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.... [2]

(b) Fig. 2.1 shows the variation with pressure p of the density ρ of a fixed mass of gas at 290 K and at another temperature T.

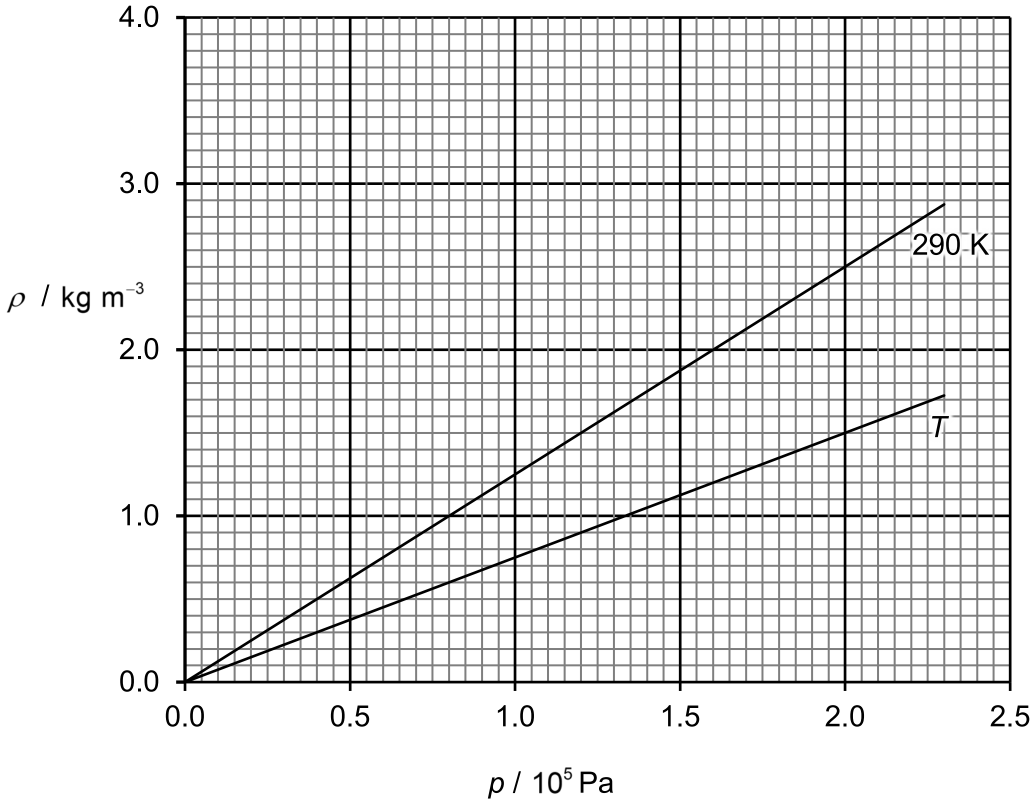

Fig. 2.1

(i) With reference to Fig. 2.1, comment on whether the gas behaves like an ideal gas.

....

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....

....

.... [2]

(ii) Determine the root-mean-square speed of the molecules of the gas at 290 K.

root-mean-square speed = ... m s^{-1} [3]

(iii) Deduce whether the temperature T is higher or lower than 290 K.
Explain your answer.

....

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....

.... [2]

3 Fig. 3.1 shows an isolated conducting sphere that has a charge of $-Q$ and radius R. The charge on the sphere may be considered to be a point charge at its centre.

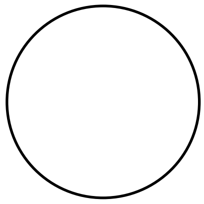

Fig. 3.1

(a) On Fig. 3.1, draw:

(i) eight field lines to represent the electric field surrounding the sphere, [1]

(ii) three equipotential lines to represent the electric potential surrounding the sphere. [1]

(b) Point X is on the surface of the sphere and point Y is $8R$ from the centre of the sphere as shown in Fig. 3.2.

An electron is initially at rest at point X.

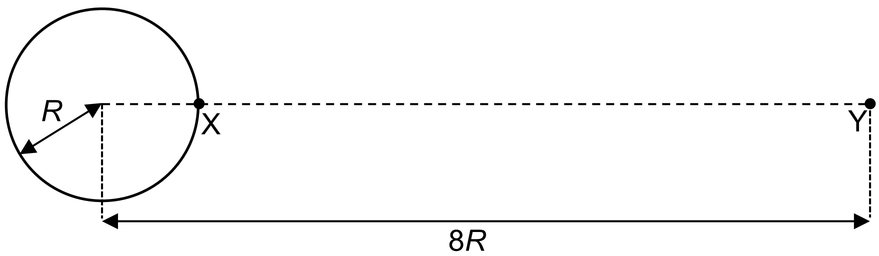

Fig. 3.2

(i) Write down an expression for the initial acceleration of the electron in terms of Q, R, elementary charge e, mass of the electron m_e and the permittivity of free space ε_0.

[1]

(ii) Describe and explain the subsequent motion of the electron.

....

....

....

....

.... [2]

(iii) The kinetic energy of the electron at point Y is 7.0×10^{-6} J.

On Fig. 3.3, draw:

1. a line, labelled P, to show how the electric potential energy of the electron varies with distance r from the centre of the sphere as it moves from R to $8R$, [2]

2. a line, labelled K, to show how the kinetic energy of the electron varies with distance r from the centre of the sphere as it moves from R to $8R$. [2]

Use the space above Fig. 3.3 for any working.

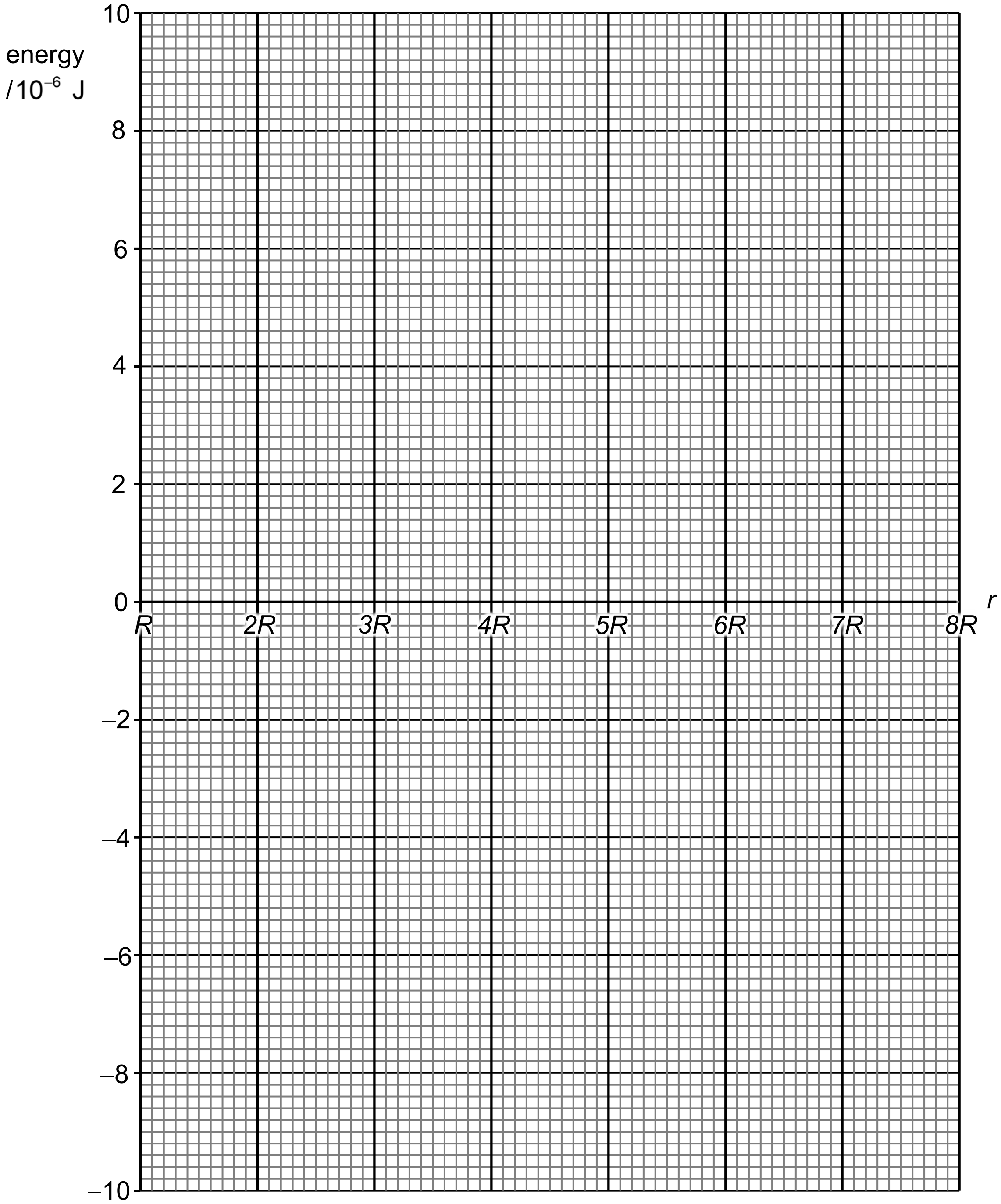

Fig. 3.3

4 Cell A of electromotive force (e.m.f.) 2.00 V and negligible internal resistance is connected to a $2.04\ \text{k}\Omega$ resistor, a variable resistor R and a uniform resistance wire PQ as shown in Fig. 4.1.

The wire PQ has length 1.20 m and resistance $20.0\ \Omega$.

A high-resistance voltmeter is connected across the $2.04\ \text{k}\Omega$ resistor.

Cell B of unknown e.m.f. E and internal resistance $0.50\ \Omega$, together with a $1.50\ \Omega$ resistor and switch S, are connected across the $2.04\ \text{k}\Omega$ resistor. G_1 is a galvanometer.

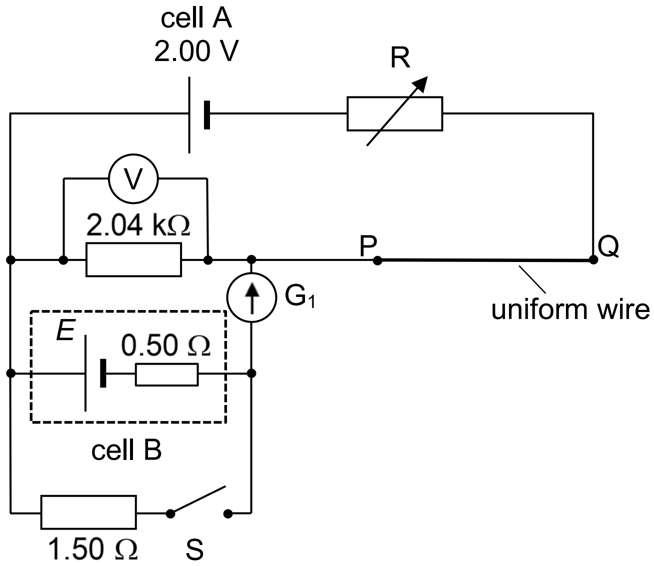

Fig. 4.1

(a) Switch S is left open. A null deflection on G_1 is achieved by adjusting the resistance of R. The voltmeter reading is 1.02 V when G_1 shows a null deflection.

(i) State the e.m.f. E of cell B.

E = ... V [1]

(ii) Determine the power supplied by cell A.

power = ... W [2]

(b) Switch S is closed and the resistance of R is adjusted until G_1 shows a null deflection again. Determine the new voltmeter reading across the 2.04 kΩ resistor.

voltmeter reading = ... V [2]

(c) To determine the e.m.f. of a thermocouple, the thermocouple and a galvanometer G_2 are added to the circuit as shown in Fig. 4.2.

With switch S closed, both G_1 and G_2 show null deflection when the sliding contact J is moved such that the length PJ is 40.0 cm.

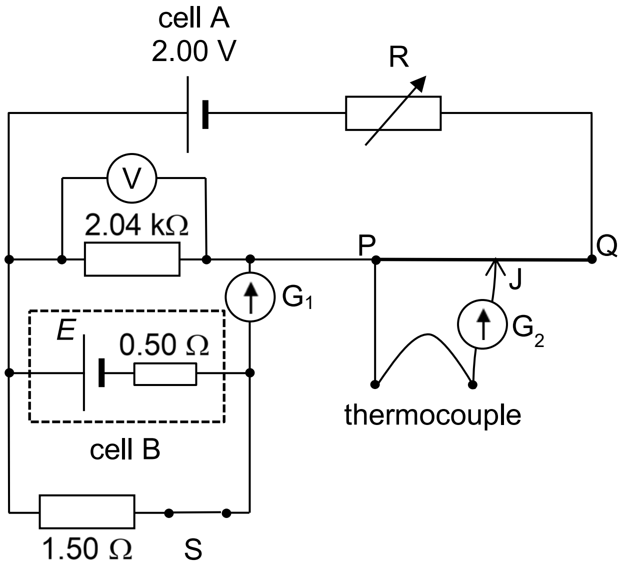

Fig. 4.2

Determine the e.m.f. of the thermocouple.

e.m.f. = ... V [3]

5 Fig. 5.1 shows a current clamp, which is an electrical device that measures the alternating current I in a wire. It consists of jaws with a soft iron core which can open and clamp around the current-carrying wire.

A coil is wrapped around the inner end of the jaws, which is connected to some electronics that measure the induced current in the coil due to the current-carrying wire's varying magnetic field.

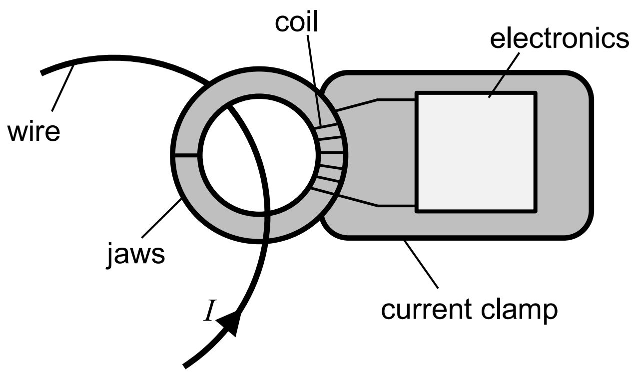

Fig. 5.1

(a) Explain why this design for a current clamp is unable to measure a direct current in the wire.

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.... [2]

(b) Suggest why using a current clamp to measure a large live current would be safer than using a digital ammeter.

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....

....

.... [2]

(c) A long straight wire carrying an alternating current I passes through the centre of the circular jaws of a current clamp as shown in Fig. 5.2.

The coil wrapped around the right side of the jaws has 9 turns and cross-sectional area $2.0\ \text{cm}^2$.

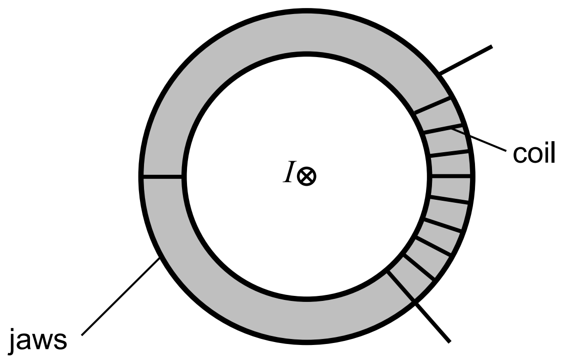

Fig. 5.2

(i) At an instant in time, the magnetic flux linkage of the coil is $0.72\ \mu\text{Wb}$.

Determine the magnitude of the average magnetic flux density B within the coil at that time.

Assume that the magnetic flux density within the coil is uniform and perpendicular to the cross-sectional area of the coil.

B = ... T [2]

(ii) Fig. 5.3 shows the variation with time t of the magnetic flux linkage Φ through the coil.

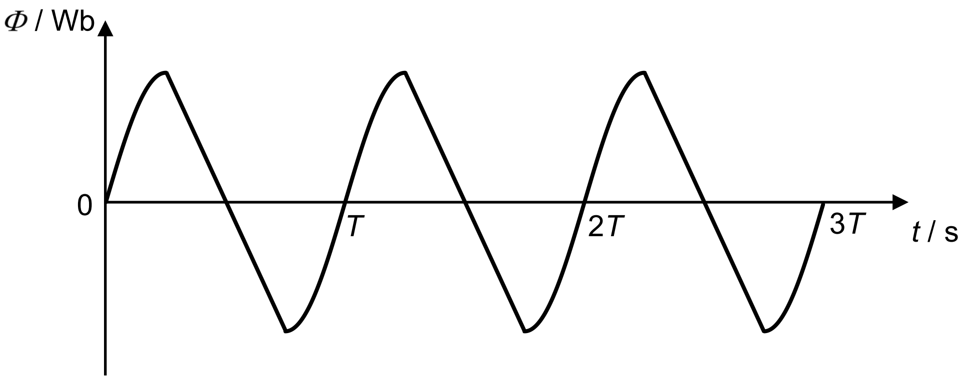

Fig. 5.3

On Fig. 5.4, sketch the variation with time t of the induced e.m.f. E in the coil from time $t = 0$ to $t = 3T$.

Label T, $2T$ and $3T$ on the horizontal axis.

Fig. 5.4

[2]

(iii) State and explain how the value of the r.m.s. current induced in the coil would change if the soft iron core in the jaws is removed.

....

....

....

....

.... [2]

6 **(a)** The photoelectric effect is studied using the set-up in Fig. 6.1. The set-up consists of a variable e.m.f. source *E* connected to a metal plate and an electrode C in an evacuated tube.

When indigo light of wavelength 450 nm is incident on the metal plate, the high-resistance voltmeter reads 1.20 V when the reading on the micro-ammeter just goes to zero.

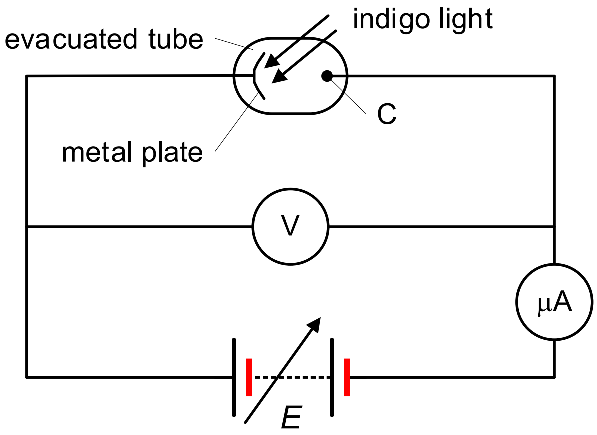

Fig. 6.1

(i) Calculate the energy of a photon of indigo light.

energy of photon = ... J [1]

(ii) State the significance of the voltmeter reading of 1.20 V.

.... [1]

(iii) Calculate the work function of the metal plate.

work function = ... J [2]

(iv) Explain why the voltmeter reading remains unchanged when indigo light of greater intensity is incident on the metal plate.

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.... [2]

(b) Fig. 6.2 shows the four lowest energy levels of the outermost electron in a mercury atom.

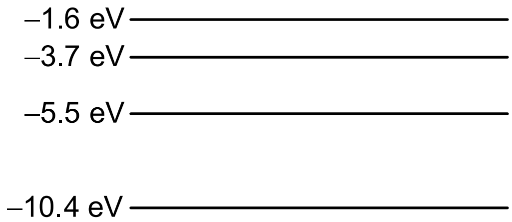

Fig. 6.2

(i) Explain why the energy of each energy level is negative.

....

....

.... [1]

(ii) Calculate the ionisation energy of a mercury atom. Explain your working.

ionisation energy = ... J [1]

(iii) Determine the wavelength of the photon emitted when an electron de-excites from the –1.6 eV energy level to the –5.5 eV energy level.

wavelength = ... m [2]

7 In the Huygens-Cassini space mission to study the planet Saturn and its system, including its rings and natural satellites, a nuclear-powered spacecraft used plutonium-238 $\left({}^{238}_{94}\text{Pu}\right)$ as the main source of energy for propulsion.

At the launch of the spacecraft, the source of plutonium-238 contained 7.20×10^{25} nuclei. Each plutonium-238 nucleus emits an α-particle of energy 5.50 MeV. The half-life of plutonium-238 is 87.7 years.

The energy of the emitted α-particles is converted into thermal energy and then into electrical energy to power the propulsion system of the spacecraft.

The mission was planned to last 20 years before the spacecraft was de-orbited to burn up in Saturn's upper atmosphere.

Calculate

(a) the activity of the plutonium source at the time of launch,

activity = ... Bq [3]

(b) the power delivered by the plutonium source at the time of launch,

power = ... W [2]

(c) the percentage of the power still available 10 years after the launch.

percentage available = ... % [2]

(d) Suggest a reason why plutonium-238 was used as a source of power for the spacecraft rather than solar panels.

....

....

.... [1]

(e) In the decay of a stationary plutonium-238 nucleus, the total energy released is 5.59 MeV. However, the energy of the emitted α-particle from the decay is 5.50 MeV.

Explain the difference in the two values.

....

....

.... [1]

8 Francis William Aston was a British physicist and chemist who discovered neon *isotopes* using a mass spectrometer. A mass spectrometer is a powerful analytical tool used to identify and quantify the composition of samples based on the mass-to-charge ratio m/z of ions, where m is the mass number and z is the charge number. For example, a hydrogen ion has a m/z ratio of 1 and an alpha-particle has a m/z ratio of 2.

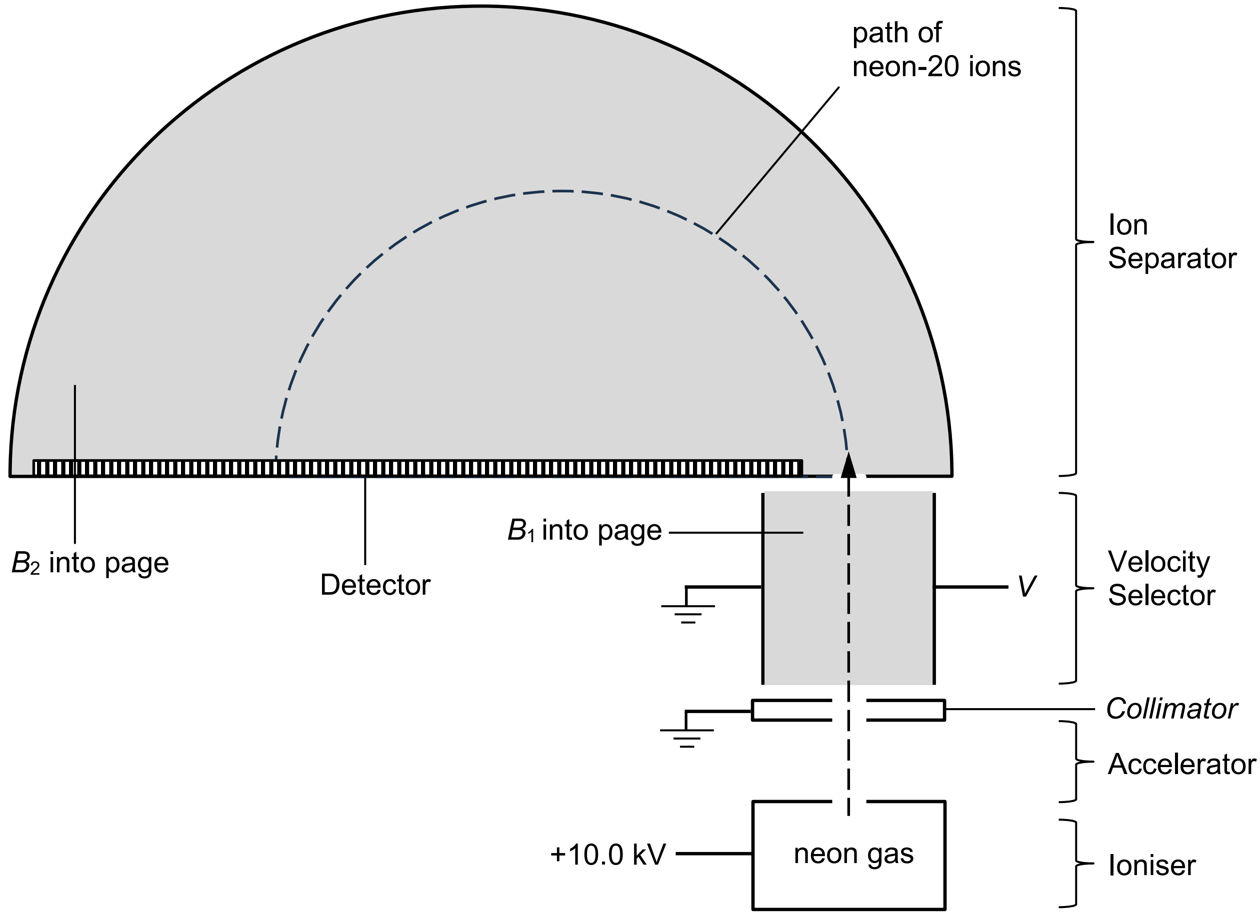

Fig. 8.1

Fig. 8.1. shows the parts of a mass spectrometer used to study the isotopes of neon gas. The purpose of each part is explained below.

1. **Ioniser:** The neon gas is ionised to form positive ions.

2. **Accelerator:** The ions are accelerated through a potential difference to gain kinetic energy.

3. **Velocity Selector:** The ions enter the region at right angles to both the electric field and the magnetic field, which has magnetic flux density B_1. Only ions of a specific velocity pass through this region undeflected.

4. **Ion Separator:** The ions enter the region at right angles to the edge of the uniform magnetic field of magnetic flux density B_2. The radii of the paths travelled by the ions depend on their m/z ratio. The detector is able to determine the number of ions incident on its sensors, producing a mass-spectrum that represents the relative abundance of ions at each m/z ratio.

Fig. 8.2 shows the relative abundance of singly charged positive ions of three isotopes of neon with their respective *m*/*z* ratio. Relative abundance refers to the ratio of the number of a particular ion to the total number of all the ions in the mass spectrum.

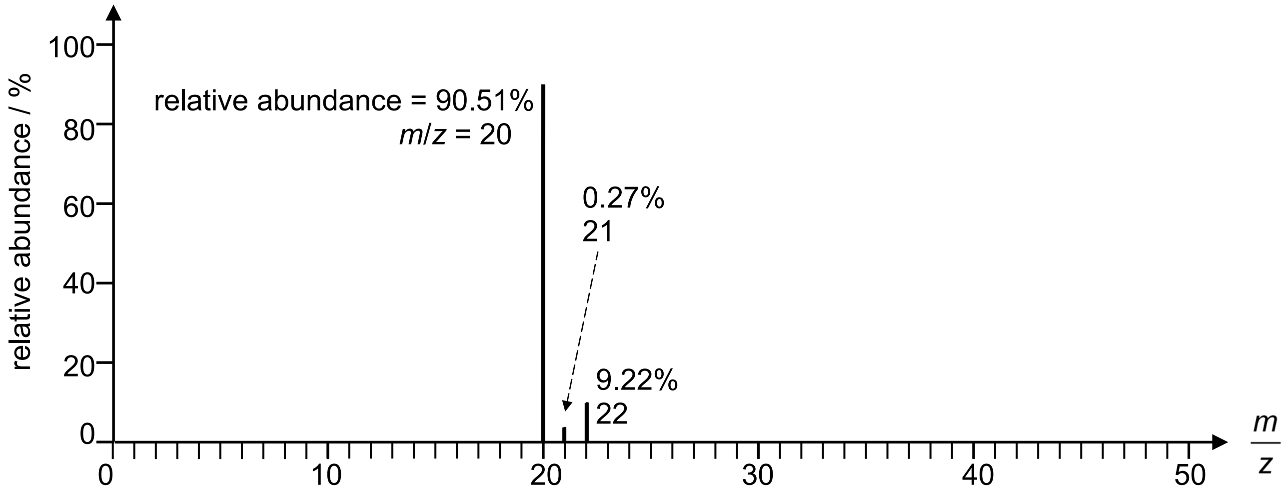

Fig. 8.2

(a) Explain what is meant by the term *isotopes*.

....

.... [1]

(b) Suggest how the gaseous neon atoms can be ionised in the Ioniser.

....

.... [1]

(c) Explain the purpose of the Collimator.

....

.... [1]

(d) Some singly charged positive neon-20 ions are formed at rest in the Ioniser.

(i) Show that these ions exit the Accelerator at a speed of 3.10×10^5 m s^{-1}.

[2]

(ii) Determine the potential V of the plate in the Velocity Selector for these ions to pass through undeflected, where $B_1 = 0.160$ T and the separation of the parallel plates is 0.100 m.

V = ... V [2]

(iii) State and explain the path within the Velocity Selector of neon-20 ions that have lost some kinetic energy due to collisions with other ions.

....

....

....

.... [2]

(iv) The path of these singly charged positive neon-20 ions is shown in Fig. 8.1.

On Fig. 8.1, draw the path of singly charged positive neon-21 ions that enter the Ion Separator at the same speed as the neon-20 ions.

[1]

(e) The relative atomic mass of an element is a unitless quantity that is defined as the ratio of the average mass of its atoms to the unified atomic mass unit, and it takes into account the mass of each isotope and its proportion in the environment.

Use Fig. 8.2 to determine the relative atomic mass of neon to 4 significant figures.

relative atomic mass = ... [2]

(f) Some of the neon isotopes that enter the Ion Separator are doubly charged positive ions.

On Fig. 8.2, indicate, using vertical lines, the positions of the doubly charged positive ions of the three isotopes of neon.

[1]

(g) State the changes that need to be made to the mass spectrometer in Fig. 8.1 when negative ions are analysed.

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....

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.... [2]

END OF SECTION B